

1 Solution — GIAM 6.3

Problem 4

1.1 Problem

The two diagrams both depict the Petersen graph: the left-hand drawing emphasizes a five-fold symmetry (with vertices already labeled A – J); the right-hand drawing emphasizes a three-fold symmetry (unlabeled). Label the right-hand drawing using A through J so that the two are exhibited as the same graph.

1.2 Setup — Edges of the Petersen Graph

The five-fold symmetric drawing (left) lays out the Petersen graph as an outer pentagon plus an inner pentagram joined by spokes:

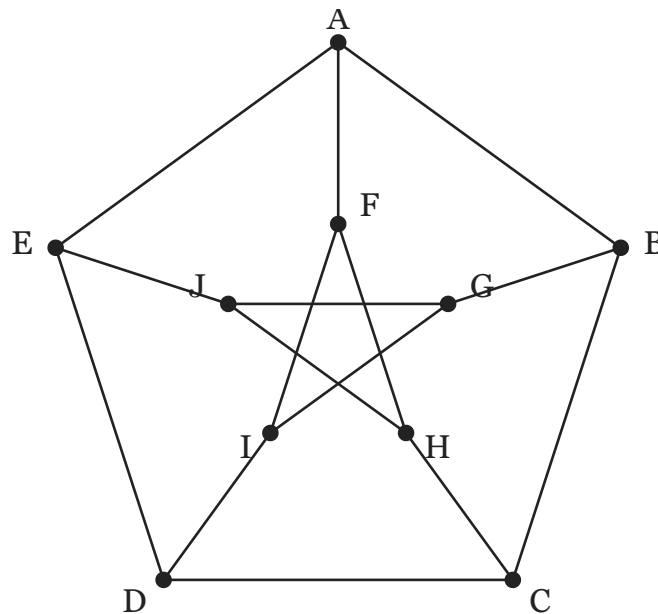


Figure 1.1: The Petersen graph drawn with five-fold symmetry — outer pentagon $ABCDE$, inner pentagram $FGHIJ$, and five radial spokes.

Reading off this drawing, the Petersen graph G has 10 vertices and the following 15 edges:

- **Outer pentagon:** AB , BC , CD , DE , EA .

- **Spokes:** AF, BG, CH, DI, EJ .
- **Inner pentagram:** FH, FI, GI, GJ, HJ .

Equivalently, the neighbor lists of all ten vertices:

$$\begin{aligned}
 A &: B, E, F \\
 B &: A, C, G \\
 C &: B, D, H \\
 D &: C, E, I \\
 E &: A, D, J \\
 F &: A, H, I \\
 G &: B, I, J \\
 H &: C, F, J \\
 I &: D, F, G \\
 J &: E, G, H
 \end{aligned}$$

Every vertex has degree **3**, as expected.

1.3 Strategy — Use the 3-fold Symmetry

The right-hand drawing has 3-fold rotational symmetry. The rotation by 120° must induce an *automorphism* of G — otherwise the symmetry would not be a true symmetry of the graph. A non-trivial planar rotation of order **3** fixes exactly one point (its center), so the induced automorphism fixes exactly one vertex; the other nine split into three orbits of three.

Consequently the right-hand drawing's vertex set partitions as:

- **1 center** vertex (the unique fixed point of the rotation),
- **3 inner triangle** vertices (one orbit),
- **6 outer hexagon** vertices (two alternating orbits).

So labeling the right-hand drawing reduces to:

1. Picking a center.
2. Identifying its three neighbors (the inner triangle).
3. Determining the cyclic order of the remaining six (the outer hexagon).

1.4 Step 1 — The Center

Take the center to be F . (Any vertex would work by overall symmetry of G ; choosing F keeps the bookkeeping tidy.)

1.5 Step 2 — The Inner Triangle

The inner triangle consists of F 's three neighbors:

$$\{A, H, I\}.$$

Note these three vertices are *pairwise non-adjacent* — they form an independent set, not a graph-theoretic triangle. The “triangle” is purely geometric (their positions in the drawing).

1.6 Step 3 — The Outer Hexagon

The remaining six vertices, $\{B, C, D, E, G, J\}$, occupy the outer hexagon. Find the cyclic order by listing each one's two *outer* neighbors — i.e. its neighbors that lie in $\{B, C, D, E, G, J\}$ (excluding the inner labels $\{A, H, I\}$ and the center F).

Vertex	All neighbors	Outer neighbors
B	A, C, G	C, G
C	B, D, H	B, D
D	C, E, I	C, E
E	A, D, J	D, J
G	B, I, J	B, J
J	E, G, H	E, G

Table 1.1

These outer-neighbor pairs trace a single 6-cycle:

$$B — C — D — E — J — G — B.$$

So the outer hexagon (in cyclic order) is B, C, D, E, J, G .

1.7 Step 4 — Inner-to-Outer Connections

Each inner-triangle vertex is adjacent to F and to two of the outer-hexagon vertices:

- A — outer neighbors B, E : positions 0 and 3 on the hexagon, i.e. *diametrically opposite*.
- H — outer neighbors C, J : positions 1 and 4, opposite.
- I — outer neighbors D, G : positions 2 and 5, opposite.

So each inner vertex sends two “long” edges to opposite hexagon corners — and these are exactly the three crossing curves visible in the right-hand drawing.

1.8 The Labeling

Assemble the pieces into the right-hand drawing:

- **Center:** F .
- **Inner triangle** (the three vertices closest to F , at 120° positions): A, H, I . Place them so that
 - A ’s two long edges meet hexagon corners B and E ,
 - H ’s long edges meet C and J ,
 - I ’s long edges meet D and G .
- **Outer hexagon** (clockwise around the boundary): B, C, D, E, J, G .

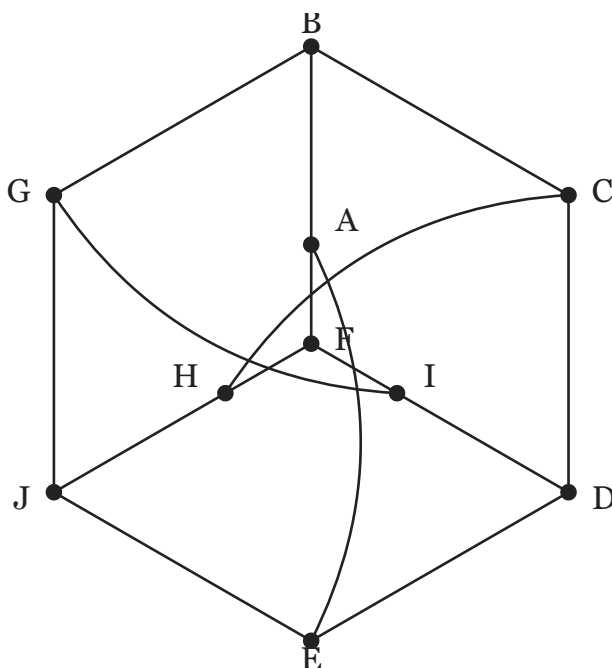


Figure 1.2: The Petersen graph drawn with three-fold symmetry: F at the centre, inner triangle $A H I$, outer hexagon $B C D E J G$ in clockwise order. The three colinear radial lines pair each inner-triangle vertex with the closer hexagon corner; the three crossing curves pair each inner-triangle vertex with the diametrically opposite corner.

In the figure above, the three short radial segments $FA-AB$, $FH-HJ$, $FI-ID$ continue along straight lines through F ; the three crossing curves AE , HC , IG each connect an inner-triangle vertex to the *opposite* hexagon corner.

Any 120° or 240° rotation of these labels — that is, cycling $A \rightarrow H \rightarrow I \rightarrow A$ together with $B \rightarrow J \rightarrow D \rightarrow B$ and $C \rightarrow G \rightarrow E \rightarrow C$ — also works, by the 3-fold symmetry of the drawing.

1.9 Verification — All 15 Edges Match

Listing the right-hand drawing's edges with the new labels:

- **Outer hexagon (6):** BC, CD, DE, EJ, JG, GB .
- **Inner spokes (3):** FA, FH, FI .
- **Long diagonals (6):** AB, AE, HC, HJ, ID, IG .

Cross-checking against the Petersen graph from the left-hand drawing:

- AB, BC, CD, DE, AE — the outer pentagon. ✓
- FA, GB, HC, ID, EJ — the spokes. ✓
- FH, FI, IG, JG, HJ — the pentagram. ✓

All 15 edges are accounted for exactly once, and no edge appears that isn't in the Petersen graph. The labeling preserves adjacency, so it is a valid graph isomorphism between the two drawings. ■

1.10 Remark — Graph Isomorphism Is an Equivalence Relation

Tying the answer back to the chapter's theme: graph isomorphism is itself an equivalence relation on the class of (simple) graphs.

- **Reflexive.** The identity map $G \rightarrow G$ is an isomorphism.

- **Symmetric.** If $\varphi : G \rightarrow H$ is an isomorphism, then $\varphi^{-1} : H \rightarrow G$ is one as well (a bijection that preserves adjacency in one direction preserves it in the other).
- **Transitive.** The composition of two isomorphisms is an isomorphism.

The two drawings in this problem are different *representatives* of the same equivalence class — the “Petersen graph” class. The labeling above is an explicit witness to this equivalence: a vertex bijection that preserves adjacency.